Workloads Cloud AI Platforms: AWS vs. GCP vs. Azure for Machine-Learning Workloads

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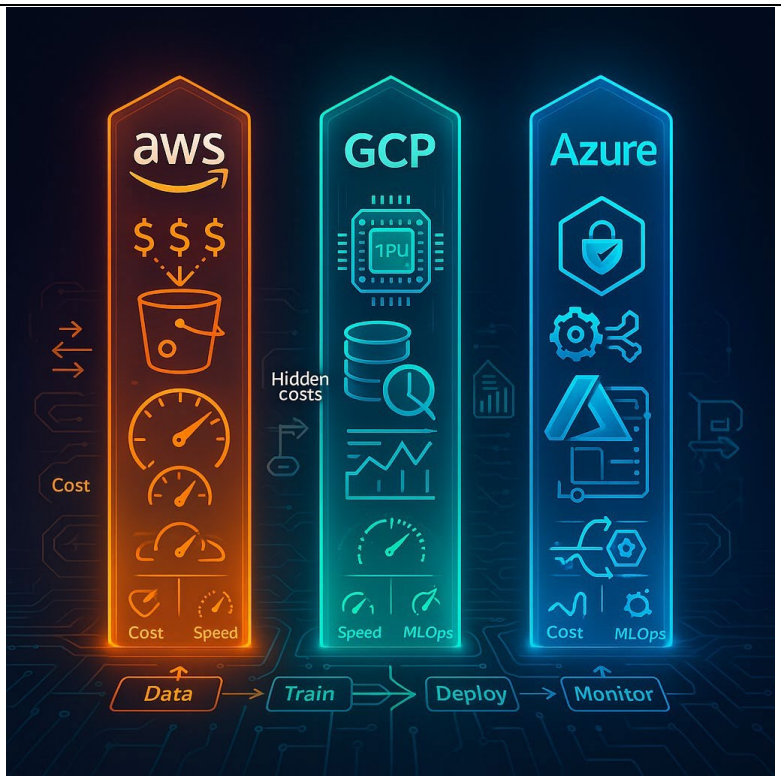
Nov 6, 2025

Audience: engineering managers, machine-learning (ML) leads, and founders, validating an ML stack.

Note: Pricing and limits here are illustrative. They change often—always check the current documentation and your region.



How we evaluate

We compare the platforms across three practical dimensions:

- **Latency & throughput** on like-for-like GPU (Graphics Processing Unit) setups
- **End-to-end cost** of training and serving across different data scales
- **MLOps (Machine Learning Operations) integration**: deployment, monitoring, and model lifecycle tooling

Quick prototyping tip: BotHub is an aggregator of AI models that hides API-key wrangling and setup, so you can focus on experiments instead of plumbing. New accounts get promo credits for test runs.

Amazon SageMaker (AWS — Amazon Web Services)

> A fully managed service covering the ML lifecycle — from data prep and experimentation to deployment, monitoring, and governance.

Pricing posture

AWS markets a **pay-for-what-you-use** model across compute, storage, and data processing, plus optional discount programs (e.g., **SageMaker Savings Plans**, introduced in 2021) and a free tier.

Gotchas the author calls out:

- **Ancillary charges** add up: CloudWatch (≈$0.10 per 1K metrics), EBS (Elastic Block Store) volumes for notebooks, and inter-AZ (Availability Zone) data transfer (~$0.01/GB). These are easy to miss in early budgets.

- **Savings Plan trade-offs: commitment is tied to specific** instance families; a quoted **$547/year** minimum can be uneconomical for small projects and hard to reallocate if your workload shifts.
- **Free tier limits: ~250 hours** on `ml.t2.medium` —about **10 days** nonstop—and **no GPU** access. Useful for clicking around the UI, not for running meaningful modern training.
- **SageMaker Studio** (≈$24/month) unlocks experiment tracking, data UX, and collaboration. Without it, the free tier feels too bare for pro use.

Net: SageMaker delivers strong cost savings **once workloads are steady and predictable**. In early-stage R&D, hidden line items can push actuals **30–50%** above the napkin math.

Flexibility & developer experience

- **Custom Docker** is first-class: bring nightly PyTorch, custom TensorFlow, etc. Expect longer image builds (reported **up to ~15 minutes** vs. 2–3 for simple bases).
- Distributed frameworks such as Ray and Dask are supported; Dask clusters often require manual setup via Processing jobs.
- Dependencies are pinned and built during image creation, helping prevent "works on my machine" drift.

Speed to production

- Commit-to-pipeline typically **takes 10–15 minutes due to mandatory Docker build steps,** even for small code changes.
- **SageMaker Projects** wires CI/CD (Continuous Integration/Continuous Delivery) via CodePipeline/CodeBuild, but you'll need some AWS familiarity.

Strength: elastic endpoints

Autoscaling for inference endpoints is excellent — smoothly handling **hundreds to tens of thousands of requests per minute** without manual tuning.

Ecosysem lock-in (the double-edged sword)

- Top-tier integrations with **S3**, **CloudWatch**, **IAM (Identity and Access Management)**, **Redshift**, and **Glue**.
- **SageMaker Pipelines** offers mature MLOps but introduces a DSL (Domain-Specific Language) and deeper integration with AWS semantics.

Google Cloud Platform (GCP) — Vertex AI

Google's **Vertex AI** favors opinionated, integrated ML development. Compared to AWS's "constructor set," Vertex AI pushes **ready-to-use cycles** and tight data integrations.

Pricing posture

- Optimized serving (e.g., via prebuilt containers) and batching can substantially reduce prediction costs; for classic vision models like ResNet-50, figures as low as ~$0.0001/request are reported in tuned setups.
- **Per-second billing** on **TPUs (Tensor Processing Units)** (e.g., **$1.35/hour** for TPU v2) reduces overpay for short tests.
- **Sustained Use Discounts:** automatic reductions after 25% monthly utilization

(e.g., **V100** dropping from ~$2.48/h to ~$1.74/h) without upfront commitments.

Hidden costs to watch:

- **Network egress/ingress:** inter-region up to **$0.19/GB**; public egress ~**$0.08/GB**. Heavy load balancing can add 15–20% to utility bills.
- **Ops telemetry:** Logging ~**$0.50/GB**; tracing/profling can rack up hundreds per month on large models.
- **Storage & analytics: BigQuery** adds ~$0.02/GB-month plus per-query costs; ML prediction queries (e.g., $0.10 per 1K) scale quickly.
- **Idle resources:** Stopped but **not fully released** GPUs/VMs can silently accrue charges — an easy $3,000/month mistake.

Operational advice: automate resource lifecycles, enforce budgets/quotas, and audit network flows. Otherwise, **network + observability** can eclipse compute by **1.5–2×**.

ML features that stand out

- **Vertex AI + BigQuery**: train on warehouse data without complex ETL (Extract, Transform, Load).
- **Feature Store** keeps training/serving features in sync; **Vertex Pipelines** gives lineage, experiment tracking, and versioning.
- **TPUs**: for large language models and vision/NLP, TPUs can significantly reduce training time (e.g., large BERT under an hour on TPUs vs. >5 hours on 8× V100 GPUs). You can **mix TPU and GPU** per stage.

Microsoft Azure Machine Learning (Azure ML)

Azure ML excels in **enterprise rollouts**: security, governance, and **hybrid** deployment are first-class. Train in the cloud, **serve on-prem** (on-premises) if needed — great for data residency and regulatory controls.

Pricing posture

- **Platform fee: ~$0.075/hour** for base ML workspace features; free tier includes **10 GB** model storage and basic monitoring.
- **GPU pricing: NVIDIA V100** around **$2.55/hour** (region-dependent).
- **Distributed CPU (Central Processing Unit):** from **$0.045 per vCPU-hour**.
- **Spot VMs** (capacity-reclaimable) can reduce compute costs by up to ~90% for preemption-tolerant jobs.

Hidden costs to watch:

- **Modular billing** across services stacks up:
- **AKS (Azure Kubernetes Service)** control plane (**~$0.10/hour**) for model serving
- **Azure Storage** (**~$0.018/GB-month**)
- **Inter-region data transfer** (**~$0.02/GB**) even on private backbone

The upside: for organizations on **Microsoft 365/Office 365**, **Active Directory**, and **Power BI**, the integration and SSO (Single Sign-On) story is hard to beat.

Enterprise strengths

- Tight hooks into **Power BI** (analytics), **Azure Data Lake** (big data), and **Defender for Cloud** (security).
- **Hybrid/edge** via **Azure IoT Edge**: deploy models to factory/retail/field gear, sync results to cloud.
- End-to-end automation: dataset versioning, lineage, model monitoring in production.

When each platform makes the most sense

Platform	Best For	Why
AWS SageMaker	Expert teams needing **fine-grained control** and **elastic at-scale inference**	Deep AWS integrations, custom Docker everywhere, robust endpoint autoscaling
GCP Vertex AI	**Research & startups** optimizing for **time-to-insight** and **warehouse-native ML**	BigQuery + Vertex AI, TPUs for rapid training, strong managed MLOps defaults
Azure ML	**Enterprises** with Microsoft stacks, **compliance,** and **hybrid/on-prem** needs	AD-driven security, Power BI, IoT Edge, predictable enterprise onboarding

A pragmatic decision framework

1. **Data gravity:** If most data lives in S3 → **SageMaker**; BigQuery → **Vertex AI**; Azure Data Lake / SQL Server → **Azure ML**.
2. **Training profile:**
   - Large-model pretraining / LLM/NLP vision → consider **TPUs** (**Vertex AI**).
   - Heavy custom containers and exotic stacks → **SageMaker**.
   - On-prem inference or strict residency → **Azure ML**.
3. **Ops maturity:**
   - Greenfeld team: prefers opinionated pipelines (**Vertex AI**).
   - AWS-savvy DevOps: **SageMaker Projects** + CodePipeline.
   - Corporate IT + SSO/compliance: **Azure ML**.
4. **Cost controls:**

- Enforce budgets, auto-shutdown, and tag-based chargeback across all clouds.
- Model serving: benchmark with and without batching; measure real egress and observability volume.

TL;DR

- **There's no universal "best."** Match the platform to your **data location**, **team skills**, **regulatory constraints**, and **latency/throughput** goals.
- **AWS SageMaker** = maximum flexibility and industrial-grade endpoint scaling — great once workloads stabilize.
- **GCP Vertex AI** = fastest path from dataset to results (BigQuery + Vertex, TPUs) with sensible managed defaults.
- **Azure ML** = enterprise integration, security, and hybrid deployments — ideal if you're already a Microsoft shop.
- Whatever you pick, **model the hidden costs** (network, logs/metrics, storage) and set up **guardrails** on day one.